ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

MECHATRONICS

PAPER 1 · PRACTICE J · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 3 Mechatronics paper 1 · practice j.

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How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Accept other correct answers and wording. Ladder programs may be written in different but correct ways: mark each required element.

QUESTION 1

1.1 The op-amp compares the voltages at its two inputs ✓ without feedback, so its gain is very high ✓; if the non-inverting (+) input is higher than the inverting (−) input, the output swings to the positive supply ✓; otherwise it swings to the negative supply / 0 V ✓. (4)

1.2 $\mathrm{V} = 5 \times 10 \div (10 + 15)$ ✓✓ $= 2\ \mathrm{V}$ ✓ (3)

1.3 It gives a steady 5 V output from a higher, varying input voltage ✓✓; the input capacitor smooths the supply and prevents oscillation ✓, and the output capacitor improves the response to sudden load changes / filters noise ✓. (4)

1.4 A small signal is applied to the base (through a coupling capacitor) ✓; the small changes in base current cause much larger changes in collector current ✓; the collector current flows through the collector resistor ✓, so a larger (inverted) voltage signal appears at the collector ✓. (4)

1.5 Any FIVE ✓✓✓✓✓: resistors; capacitors; diodes; transistors; integrated circuits (op-amps, timers, microcontrollers); relays; LEDs; voltage regulators; sensors. (5)

[20]

QUESTION 2

2.1 An incremental encoder gives a pulse for each small movement, so the controller counts pulses to work out position, and the position is lost if power fails ✓✓. An absolute encoder gives a unique code for every shaft position, so the exact position is known at once, even after a power failure ✓✓. (4)

2.2 $\mathrm{P} = 2\pi\mathrm{NT} \div 60$ ✓ $= 2\pi \times 1\,500 \times 8 \div 60$ ✓ $= 1\,256{,}64\ \mathrm{W}\ (1{,}26\ \mathrm{kW})$ ✓ (3)

2.3 Any THREE ✓✓✓: torque required; speed range; accuracy / position control (feedback); size and mass; supply voltage; cost; duty cycle. (3)

2.4 A control system that combines continuous (analogue) control with discrete (on/off, sequence) control ✓✓ – also called mixed control. Example ✓✓: a filling machine that controls the fill level continuously while a PLC runs the on/off sequence of conveyors and valves; a robot with continuous servo control and a sequence of discrete tasks. (4)

2.5 A program is the finished set of instructions that a controller or computer carries out ✓✓; programming is the process of designing, writing, testing and correcting those instructions ✓✓. (4)

2.6 Any TWO ✓✓: closed-loop feedback, so it does not lose steps; higher speed; better torque at high speed; smoother motion; more efficient. (2)

[20]

QUESTION 3

3.1 $F = p \times A = 8 \times 10^6 \times 0{,}005$ ✓✓ $= 40\,000$ N $(40$ kN$)$ ✓ (3)

3.2 $v = Q \div A = 0{,}0005 \div 0{,}005$ ✓✓ $= 0{,}1$ m/s ✓ (3)

3.3 An accumulator stores oil under high pressure even when the pump is switched off ✓; if a line is opened, this oil can be released suddenly ✓, causing injury (oil injection) or unexpected movement of actuators ✓. (3)

3.4 A simple throttle valve gives a flow that changes when the load pressure changes ✓✓; a pressure-compensated valve keeps the flow (and therefore the actuator speed) constant even when the load pressure changes ✓✓. (4)

3.5 Any FOUR ✓✓✓✓: switch off and lock out the pump; release pressure (including accumulators); lower or support loads; never check for leaks with the hand; wear safety glasses and gloves; use correctly rated hoses and fittings; clean up spills. (4)

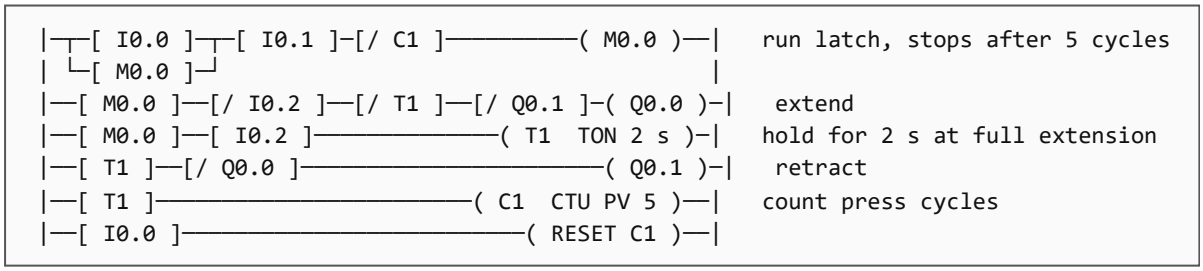
3.6 Any THREE ✓✓✓: stores the oil; cools the oil; allows air bubbles to escape; lets dirt and water settle; provides space for oil expansion; mounting point for the pump and motor. (3)

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QUESTION 4

4.1 Each time the count input changes from off to on, the accumulated value increases by 1 ✓; when the accumulated value reaches the preset, the counter's done bit switches on ✓; a reset input sets the count back to 0 ✓. (3)

4.2 Run latch: I0.0 with M0.0 holding contact ✓✓, through I0.1 (STOP) ✓ and an NC counter-done contact ✓; extend rung: M0.0 and NC I0.2 / NC T1 driving Q0.0 ✓; timer rung: I0.2 starts T1 = 2 s ✓✓; retract rung: T1 done drives Q0.1 ✓; counter rung: T1 done (or I0.2) counts C1 with preset 5 ✓✓; counter reset (START / reset input) ✓; Q0.0 and Q0.1 interlocked ✓; correct ladder symbols ✓, e.g. (13)



4.3 Any FOUR ✓✓✓✓: program changes without rewiring; built-in timers and counters; less wiring and smaller panels; easy fault-finding with status LEDs and online monitoring; more reliable (no relay contacts to wear); can connect to HMIs and networks; can store several programs / recipes. (4)

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QUESTION 5

5.1.1 Repairs are done only after a machine breaks down ✓; no planned work, but unplanned downtime and higher costs ✓. (2)

5.1.2 Maintenance is planned at fixed intervals (time or running hours) ✓ to prevent breakdowns, e.g. changing filters and oil ✓. (2)

5.1.3 The condition of the machine is measured (vibration, temperature, oil analysis) ✓ and maintenance is done only when the measurements show that a fault is developing ✓. (2)

5.1.4 A company-wide approach in which operators and maintenance staff share responsibility for the machines ✓ (daily cleaning, inspection, lubrication by operators) to achieve maximum equipment effectiveness ✓. (2)

5.2 To check the condition of the oil and the system without dismantling it ✓; the sample shows contamination (dirt, water) ✓, wear metals from pumps and cylinders ✓, and whether the oil has broken down (viscosity, additives), so maintenance can be planned before failure ✓. (4)

5.3 Any SIX in order ✓✓✓✓✓✓: isolate and lock out; release the pressure and support any load; disconnect and plug the hoses; remove the cylinder (or the end cap / gland); withdraw the piston and rod; remove the old seals and clean the grooves without scratching; fit the correct new seals, lubricated with clean oil; reassemble and torque the fasteners; refit, bleed the air and test for leaks and smooth movement. (6)

5.4 Any TWO ✓✓: date and time; machine name / number; description of the fault; cause found; action taken / parts replaced; downtime; name of the technician. (2)

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TOTAL: 100

END OF MARKING GUIDELINE